Handheld Analog Metal Detector

User Manual

Team Prometheus

1.Product Overview

Device Model: Version 1.1

Development Information

- Developed By: Team Prometheus
- Date of Manufacture: 08/12/2025
- Contact/Support: prometheusmetaldetectors@gmail.com

Description

This handheld analog metal detector locates metallic objects by converting search-coil frequency shifts into audible signals. The device utilizes a fully analog circuit design without digital processing, ensuring real-time response.

Intended Use

- Suitable For : Close-range security screening and handheld personnel inspection
  Educational demonstrations of analog electronics and electromagnetism.
- Target Environment: Security access points, Educational settings
- Operating Conditions: Dry, non-condensing environments.
- User Proficiency: Beginner.

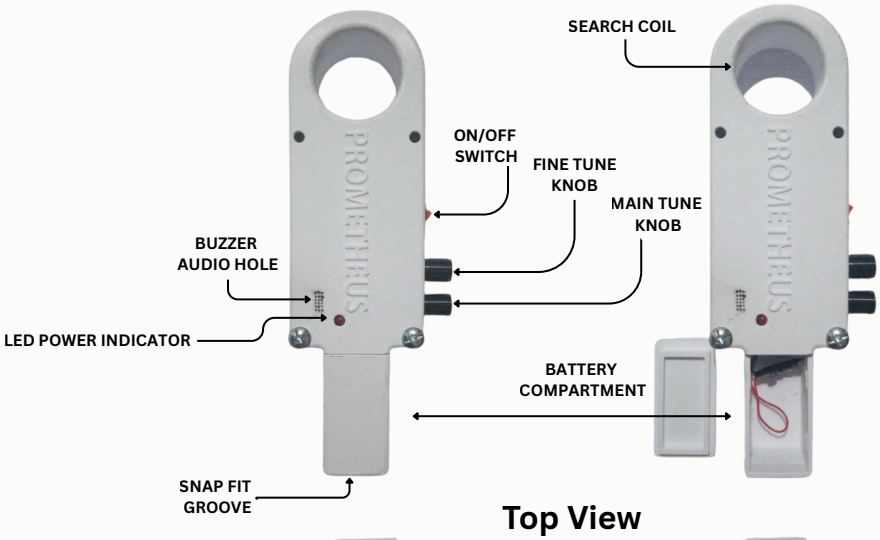
Key Specifications

- Power Source: 9V Battery.
- Operating Duration: Robust for up to 3 Hour continuous use
- Audio Output: Built in buzzer.

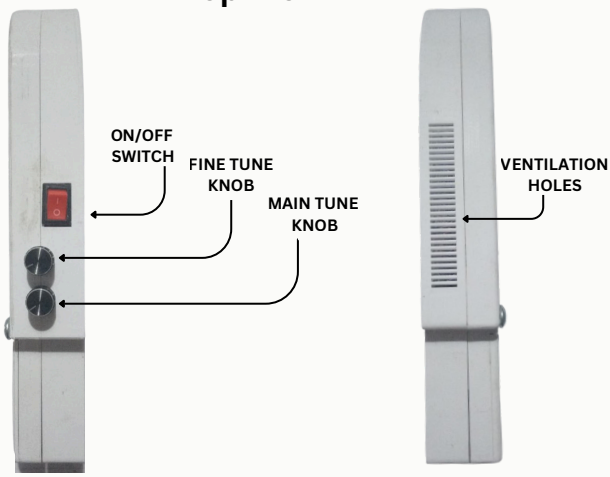
2. What is in the box

- Analog Metal Detector Unit
- User Manual
- Warranty Card
- Datasheet
- Calibrator knob replacements

3. Product Layout



Top View



Side View

- ON/OFF Switch: Turns the device power ON or OFF.
- Low-Frequency Adjustment: Controls sensitivity to slow movements.
- High-Frequency Adjustment: Controls sensitivity to fast movements.

- Audio Indicator: The buzzer increases the frequency when a metal is detected.
- LED power indicator: Indicates that the device is powered ON.

4. How to Use

4.1 Startup Procedure

- Connect a 9 V power source.
- Keep the search coil away from metal objects.
- Power ON the circuit.
- Adjust main tune knob until a steady, low-frequency tone is heard.
- Adjust fine tune knob until no sound is heard.

The detector is now ready for use.

For the battery:

- Use 9V standard DC cells. (both single charge or rechargeable are compatible
- Press the snap-fit groove in the bottom to lift up the battery compartment lid. Then insert/change your battery. Be mindful of the orientation.

4.2 Calibration Guidelines

- Perform calibration away from large metal objects.
- Re-adjust the threshold if ambient temperature changes significantly.
- If the tone drifts slowly, fine-tune the knob.

4.3 Normal Operation

- Move the search coil slowly over the target area.
- When metal is detected:
  - The buzzer tone will increase in frequency.
  - The amount of change depends on object size, distance, and material.

4.4 Notes

- Ferrous and non-ferrous metals both affect the coil but may produce different tone behaviors.
- Sudden jerky movements may cause false indications.

5. Safety and Handling

- Do not short-circuit the power supply terminals.
- Do not exceed the specified input voltage rating.
- Avoid touching the PCB while the device is powered, as this may introduce noise and affect proper operation.
- Disconnect and remove the battery if the device will not be used for an extended period.
- Protect the device from environmental factors such as dust, moisture, and direct sunlight.
- Ensure the search coil cable is properly strain-relieved to prevent mechanical damage or intermittent connections.

6. Troubleshooting

Problem	Possible Cause	Recommended Action
Device does not power ON	Battery disconnected or discharged	Check battery connection or replace battery
No audio output	Buzzer not connected or faulty	Check buzzer wiring and connections
No audio output	5 V regulator not working	Measure regulator output; verify 9 V input
Constant loud tone	Threshold set too high	Adjust main tune knob
Very weak or no response to metal	Sensitivity set too low	Increase sensitivity control knob
Detector triggers continuously	Nearby metal objects	Move to a metal-free area and recalibrate
Unstable or noisy tone	Poor grounding or environmental interference	Check ground connections and cable shielding
Tone drifts over time	Temperature or supply voltage variation	Re-adjust threshold control
False detection	Coil cable movement	Secure coil cable and improve strain relief
Limited detection range	Coil damaged or incorrectly connected	Inspect coil continuity and wiring

NOTE: Always refer technical assistance from a professional for damaged equipments

7. technical specifications

Handheld Metal Detector		
System Parameters	**The Smallest Detected Metals**	Coin-sized metals
	Adjustable Sensitivity	3 levels
	Configuration Method	Manual controlling
	Power Supply	Standard 9 V (6F22) battery
	Power Consumption	< 60mW(normal operation)
	Operating temperature	$-20°C - 55°C$
	Operating Humidity	10%–95%, non-condensing
	Protective Level	IP01
	Operating Frequency	26KHz
	Output	Buzzer output
	Duration	Continuous work for 3 h
	Net Weight	300 g (0.66 lb) (without battery)
Dimensions	**Overall dimensions**	(mm) 2600(H) × 1000(W) × 450(D)
	Passageway dimension	(mm) 2600(H) × 8000(W) × 450(D)

- For support or further inquiries, refer to team Prometheus

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